and

B is the beam of the ship.

Alternatively, the heeling moment may be calculated assuming the passengers are distributed with 4 persons per square metre on available deck areas towards one side of the ship on the decks where muster stations are located and in such a way that they produce the most adverse heeling moment. In doing so, a weight of 75 kg per passenger is to be assumed.

4.1.2 Mwind is the maximum assumed wind force acting in a damage situation:

Mwind = (P · A · Z) / 9,806 (tm)

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SOURCE: MSC.216%2882%29.pdf
is to be assumed.

4.1.2 Mwind is the maximum assumed wind force acting in a damage situation:

Mwind = (P · A · Z) / 9,806 (tm)

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where:

P = 120 N/m 2;

A = projected lateral area above waterline;

Z = distance from centre of lateral projected area above waterline to T/2; and

T = ships draught, di.

4.1.3 MSurvivalcraft is the maximum assumed heeling moment due to the launching of all fully loaded davit-launched survival craft on one side of the ship. It shall be calculated using the following assumptions:

.1 all lifeboats and rescue boats fitted on the side to which the ship has heeled after having sustained damage sha ll be assumed to be swung out fully loaded and ready for lowering;

.2 for lifeboats which are arranged to be launched fully loaded from the

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after having sustained damage sha ll be assumed to be swung out fully loaded and ready for lowering;

.2 for lifeboats which are arranged to be launched fully loaded from the stowed position, the maximum heeli ng moment during launching shall be taken;

.3 a fully loaded davit-launched liferaft attached to each davit on the side to which the ship has heeled after having sustained damage shall be assumed to be swung out ready for lowering;

.4 persons not in the life-saving appl iances which are swung out shall not provide either additional heeling or righting moment; and

.5 life-saving appliances on the side of the ship opposite to the side to which the ship has heeled shall be assumed to be in a stowed position.

5 Unsymmetrical flooding is to be kept to a minimum consistent with the efficient arrangements. Where it is necessary to corr ect large angles of heel, the means adopted

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5 Unsymmetrical flooding is to be kept to a minimum consistent with the efficient arrangements. Where it is necessary to corr ect large angles of heel, the means adopted shall, where practicable, be self-acting, but in any case where controls to equalization devices are provided they shall be operable from above the bulkhead deck. These fittings together with their controls shall be a cceptable to the Administration. Suitable information concerning the use of equalization devices shall be supplie d to the master of the ship.

5.1 Tanks and compartments taking part in such equalization shall be fitted with air pipes or equivalent means of sufficient cross-se ction to ensure that the flow of water into the equalization compartments is not delayed.

5.2 In all cases, si is to be taken as zero in those cases where the final waterline, taking into account sinkage, heel and trim, immerses:

.1 the lower edge of openings through which progressive flooding may take

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into account sinkage, heel and trim, immerses:

.1 the lower edge of openings through which progressive flooding may take place and such flooding is not accounted for in the calculation of factor si. Such openings shall include air-pipes, ventilators and openings which are closed by means of weathertight doors or hatch covers; and

.2 any part of the bulkhead deck in passenger ships considered a horizontal evacuation route for compliance with chapter II-2.

5.3 The factor si is to be taken as zero if, taking into account sinkage, heel and trim, any of the following occur in any intermediate stage or in the final stage of flooding:

.1 immersion of any vertic al escape hatch in the bulkhead deck intended for compliance with chapter II-2;

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.1 immersion of any vertic al escape hatch in the bulkhead deck intended for compliance with chapter II-2;

.2 any controls intended for the opera tion of watertight doors, equalization devices, valves on piping or on ventilation ducts intended to maintain the integrity of watertight bulkheads fr om above the bulkhead deck become inaccessible or inoperable;

.3 immersion of any part of piping or ventilation ducts carried through a watertight boundary that is located within any compartment included in damage cases contributing to the attained index A, if not fitted with watertight means of closure at each boundary.

5.4 However, where compartments assume d flooded due to prog ressive flooding are taken into account in the damage stability calculations multiple values of sintermediate,i may be calculated assuming equalization in additional flooding phases.

5.5 Except as provided in paragraph 5.3.1, openings closed by means of watertight

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be calculated assuming equalization in additional flooding phases.

5.5 Except as provided in paragraph 5.3.1, openings closed by means of watertight manhole covers and flush scuttles, small wate rtight hatch covers, remotely operated sliding watertight doors, side scuttles of the non-opening type as well as watertight access doors and hatch covers required to be kept closed at sea need not be considered.

6 Where horizontal watertight boundaries are fitted above the waterline under consideration the s-value calculated for the lower compartment or group of compartments shall be obtained by multiplying the value as determined in paragraph 1.1 by the reduction factor vm according to paragraph 6.1, which represents the probability that the spaces above the horizontal subdivision will not be flooded.

6.1 The factor vm shall be obtained from the formula:

vm = v(Hj, n, m, d) v(Hj, n, m-1, d)

where:

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spaces above the horizontal subdivision will not be flooded.

6.1 The factor vm shall be obtained from the formula:

vm = v(Hj, n, m, d) v(Hj, n, m-1, d)

where:

Hj, n, m is the least height above the baselin e, in metres, within the longitudinal range of x1(j)...x2(j+n-1) of the mth horizontal boundary which is assumed to limit the vertical extent of flooding for the damaged compartments under consideration;

Hj, n, m-1 is the least height above the baselin e, in metres, within the longitudinal range of x1(j)...x2(j+n-1) of the (m-1)th horizontal boundary which is assumed to limit the vertical extent of flooding for the damaged compartments under consideration;

j signifies the aft terminal of the damaged compartments under consideration;

m represents each horizontal boundary counted upwards from the waterline under consideration;
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